

Surd rules



1 Are the following statements true or false?

a $2 - \sqrt{2} = \sqrt{2}$

b $\sqrt{3^2} = (\sqrt{3})^2$

2 Rewrite each of the following expressions in the form $\sqrt{a}$:

a $6\sqrt{7}$

b $4\sqrt{14}$

c $3\sqrt{5}$

d $10\sqrt{10}$

3 Simplify:

a $\frac{12 + 9\sqrt{5}}{3}$

b $\sqrt{25 \times 2}$

c $(\sqrt{45})^2$

d $(7\sqrt{15})^2$

e $\sqrt{36x}$

f $\frac{\sqrt{33}}{\sqrt{11}}$

g $\sqrt{\left(1\frac{7}{16}\right)}$

Multiplication and division of surds

4 Simplify the following:

a $\sqrt{2} \times \sqrt{11}$

b $2\sqrt{2} \times 2$

c $4\sqrt{35} \times \sqrt{5}$

d $5\sqrt{7} \times 3\sqrt{5}$

e $\sqrt{5} \times \sqrt{2} \times \sqrt{7}$

f $\sqrt{343} \times \sqrt{75}$

g $\sqrt{5} \times \sqrt{7}$

h $8 \times 10\sqrt{5}$

i $\sqrt{7} \times \sqrt{3} \times \sqrt{11}$

j $\sqrt{55} \times \sqrt{11}$

k $4\sqrt{11} \times 5$

l $2\sqrt{5} \times 15\sqrt{11}$

m $7\sqrt{22} \times \sqrt{2}$

n $\sqrt{180} \times \sqrt{48}$

o $8\sqrt{15} \times 8\sqrt{5}$

p $5\sqrt{17} \times 8\sqrt{3}$

q $17\sqrt{35} \times 4\sqrt{5}$

r $8\sqrt{51} \times 9\sqrt{3}$

5 Simplify the following:

a $\sqrt{85} \div \sqrt{17}$

b $18\sqrt{7} \div 9$

c $24\sqrt{112} \div 3\sqrt{7}$

d $8\sqrt{77} \div \sqrt{7}$

e $20\sqrt{14} \div 5\sqrt{2}$

f $\sqrt{15} \div \sqrt{5}$

g $\sqrt{55} \div \sqrt{5}$

h $\sqrt{51} \div \sqrt{17}$

i $\sqrt{21} \div \sqrt{3}$

j $\sqrt{91} \div \sqrt{7}$

k $40\sqrt{7} \div 8$

l $10\sqrt{55} \div \sqrt{11}$

m $15\sqrt{22} \div \sqrt{11}$

n $4\sqrt{35} \div 2\sqrt{5}$

o $\sqrt{27} \div \sqrt{3}$

p $3\sqrt{20} \div \sqrt{5}$

q $5\sqrt{8} \div \sqrt{2}$

r $40\sqrt{96} \div 10\sqrt{6}$

s $50\sqrt{24} \div 10\sqrt{6}$

t $\sqrt{25} \div \sqrt{81}$

u $\sqrt{162} \div \sqrt{8}$

6 Simplify the following:



a $\sqrt{\frac{28}{7}}$

b $\sqrt{\frac{9}{45}}$

c $\sqrt{\frac{64}{4}}$

d $\sqrt{\frac{48}{144}}$

e $\frac{\sqrt{12}}{\sqrt{36}}$

f $\frac{\sqrt{56}}{\sqrt{14}}$

g $\frac{\sqrt{36}}{\sqrt{81}}$

h $\frac{\sqrt{72}}{\sqrt{32}}$

i $\frac{17\sqrt{21}}{\sqrt{3}}$

Mixed operations with surds

7 Simplify the following:

a $8\sqrt{3} + 18\sqrt{3}$

b $-12\sqrt{5} + 6\sqrt{5} - 12\sqrt{5}$

c $8\sqrt{6} + 17\sqrt{13} + 19\sqrt{6} - 6\sqrt{13}$

d $17\sqrt{6} - 8\sqrt{11} + 3\sqrt{6} + 13\sqrt{11}$

e $\sqrt{6} + \sqrt{54}$

f $\sqrt{180} + \sqrt{500}$

g $\sqrt{32} - \sqrt{98}$

h $4\sqrt{45} + 5\sqrt{180}$

8 Use the formula $\sqrt{a} + \sqrt{b} = \sqrt{a + b + 2\sqrt{ab}}$ to simplify the following:

a $\sqrt{3} + \sqrt{12}$

b $\sqrt{8} + \sqrt{2}$

c $\sqrt{18} + \sqrt{8}$

d $2\sqrt{20} + 2\sqrt{5}$

9 Expand and simplify:

a $\sqrt{7}(-5 + \sqrt{5})$

b $7\sqrt{2}(\sqrt{7} + 6)$

c $7\sqrt{11}(-6\sqrt{2} + \sqrt{7})$

d $(\sqrt{13} + 5)(\sqrt{2} - 3)$

e $(\sqrt{7} - \sqrt{13})(\sqrt{2} - \sqrt{11})$

f $(\sqrt{11} + 1)^2$

g $(2\sqrt{5} + 3)^2$

h $\sqrt{11}(\sqrt{7} + 4)$

i $\sqrt{7}(3 + \sqrt{3})$

j $\sqrt{2}(\sqrt{11} - 6)$

k $3\sqrt{3}(\sqrt{13} - 5)$

l $\sqrt{3}(\sqrt{11} + \sqrt{13})$

m $4\sqrt{7}(\sqrt{2} - \sqrt{11})$

n $3\sqrt{5}(\sqrt{55} + \sqrt{11})$

o $8\sqrt{2}(\sqrt{3} - 3\sqrt{7})$

p $5\sqrt{2}(3\sqrt{5} + 4\sqrt{7})$

q $7\sqrt{3}(\sqrt{15} + \sqrt{60})$

r $11\sqrt{3}(3\sqrt{5} - \sqrt{20})$

s $8\sqrt{11}(3\sqrt{7} - 4\sqrt{5})$

10 Simplify the following:



a $3\sqrt{5} \times 8\sqrt{10} \div 2\sqrt{2}$

b $\frac{8\sqrt{12} \times 3\sqrt{2}}{4\sqrt{10}}$

c $\frac{\sqrt{15} + \sqrt{21}}{\sqrt{3}}$

d $\frac{3\sqrt{2} \times 2\sqrt{6}}{\sqrt{8} \times \sqrt{20}}$

11 Find the values of x and y in the following equations:

a $(8\sqrt{91} - 5\sqrt{7})^2 = x + y\sqrt{13}$

b $(\sqrt{x} + y)^2 = 14 - 6\sqrt{5}$

12 Consider the expression $(\sqrt{13} + 4)^2 + (\sqrt{13} + m)^2$.

a Expand and simplify the expression.

b Hence, what value of m can be substituted into $(\sqrt{13} + 4)^2 + (\sqrt{13} + m)^2$ so that it has a rational value?

Rationalise the denominator

13 State whether the following results in a rational number:

a $(\sqrt{8} + \sqrt{3}) \times (\sqrt{8} + \sqrt{3})$

b $(\sqrt{8} + \sqrt{3}) \times (\sqrt{8} - \sqrt{3})$

c $(-(\sqrt{8} + 2)) \times \sqrt{8}$

14 Rationalise the denominator of the following:

a $\frac{1}{\sqrt{11}}$

b $\frac{\sqrt{2} - 2}{\sqrt{11}}$

c $\frac{7}{\sqrt{12}}$

d $\frac{14\sqrt{11}}{9\sqrt{13}}$

15 Express the fractions in simplest form with a rational denominator:

a $\frac{\sqrt{5}}{\sqrt{15}}$

b $\frac{7\sqrt{30}}{\sqrt{10}}$

c $\frac{9}{\sqrt{5} - 7}$

d $\frac{8}{9\sqrt{11} - 5}$

e $\frac{\sqrt{7} + \sqrt{2}}{\sqrt{7} - \sqrt{2}}$

f $(\sqrt{10} + 1)^{-1}$

g $\frac{28}{\sqrt{10} - \sqrt{3}}$

h $\frac{\sqrt{7}}{5\sqrt{6} + 4\sqrt{2}}$

16 Express the following in simplest form with rational denominator:

a $\frac{1}{\sqrt{2} + 5} + \frac{1}{\sqrt{2} - 5}$

b $\frac{3}{4\sqrt{5}} - \frac{3}{2\sqrt{2}}$

17 Show that the following expression is a rational number:

$$\frac{3}{\sqrt{2}+3} + \frac{3}{\sqrt{2}-3}$$

18 If $x = 2 + \sqrt{3}$, simplify and rationalise the denominator of the following expressions:

a $\frac{x - \sqrt{3}}{x + \sqrt{3}}$

b $\left(x + \frac{1}{x}\right)^2$

19 If $x = 4 + \sqrt{3}$, simplify and rationalise the denominator of the following expressions:

a x^2

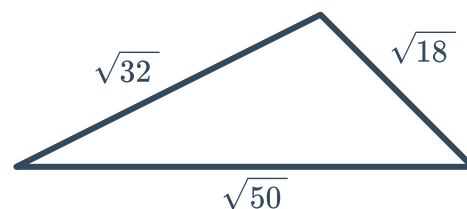
b $\frac{x+1}{x-1}$

20 Express the following as a single surd:

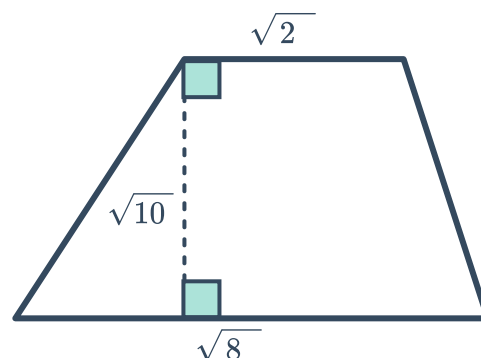
$$42 + \frac{6}{7 + 5\sqrt{2}}$$

Applications

21 Find the perimeter of the triangle in simplified surd form:



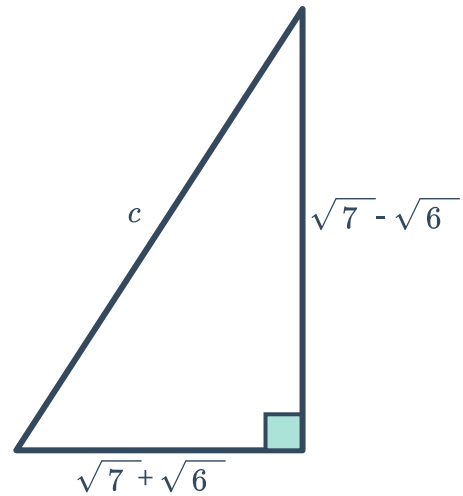
22 Find the area of the trapezium in surd form:



23 A rectangle has a width of $(\sqrt{7} - 2)$ m and a length of $(\sqrt{7} + 2)$ m.

- a** Find the exact perimeter of the rectangle.
- b** Find the exact area of the rectangle.

24 Find c , the length of the hypotenuse of the triangle below. Leave your answer in surd form.



25 Find the exact perpendicular height of a triangle whose area is $36\sqrt{15}$ square centimetres and whose base measures $8\sqrt{5}$ centimetres.

26 A rectangle with height h and width w are believed to be more aesthetically pleasing if their lengths are in the following ratio:

$$\frac{w}{h} = \frac{2}{\sqrt{5} - 1}$$

- a** Rationalise the denominator of the ratio.
- b** Find a decimal approximation for this ratio correct to the nearest thousandth.

27 A rectangle has a length of $(\sqrt{10} + 3)$ meters and an area of 4 square meters. What is the exact width of this rectangle? Give your answer with a rational denominator.